OPEN STEM PROJECT

Engineering Design and Development (EDD)
Martin Luther King High School
Team Name: M.O.J.O.

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Element L

Reflection on the Design Project

1 PROBLEM STATEMENT

Commercially available STEM toys designed for children—particularly those marketed towards girls—often fail to sustain inquiry-driven engagement through narrower technical content, limited opportunities for authentic engineering practices, and inconsistent value relative to gender-neutral alternatives, reducing opportunities for meaningful guardian-child co-play and dampening long-term interest in STEM among children ages 6–12. The M.O.J.O. team developed a modular rover-based construction kit as a proposed solution and now presents recommendations for prototype improvement and a forward-looking dream design concept.

2 INTRODUCTION

Given additional time and resources, the M.O.J.O. team would pursue two parallel tracks of development: near-term refinement of the R3 prototype to address the specific failure modes and gaps identified through formal testing and external evaluation, and a more ambitious long-term redesign that reimagines the platform at a larger scale with substantially upgraded hardware. The most immediate priority would be resolving the battery module retention failure identified in Element I drop testing, as this is the only structural weakness that produced a formal test failure and represents an actionable gap between the current prototype and full R3 specification compliance. Beyond that fix, the team would focus on realizing the line-following brain module—the one design requirement that was documented but not physically implemented in R3—as it is the most educationally significant unfinished feature of the platform. In the longer term, the dream design concept envisions a four-wheel-drive rover at a significantly larger form factor, with stronger motors, a higher-capacity off-the-shelf battery connected via USB-C, a purpose-built custom PCB brain replacing the current breadboard-and-development-board architecture, and a chassis aesthetic inspired by planetary exploration rovers. This direction is grounded in stakeholder feedback from the Element J showcase, where respondents noted that the 3D-printed appearance of the current prototype raised questions about perceived durability and product finish, and where multiple respondents specifically praised the innovation and novelty of the technology—suggesting appetite for a more visually distinctive and capable platform. The dream design is not a revision of

R3 but a generational step forward that would require a substantially different engineering and manufacturing approach.

3 Improvements and Current Prototype

The following improvements address the R3 prototype directly, grounded in test results from Element I and stakeholder feedback from Element J.

Battery module retention reinforcement. The clip-based interface between the battery module and chassis was identified as the primary failure mode in Element I drop testing, producing module ejection under top-down and lateral impact in multiple runs. The improvement would introduce a secondary locking mechanism—either a keyed friction-fit deepening, a positive-retention tab requiring deliberate release, or a complementary clip on the opposing face of the joint. This directly addresses the R3 drop resilience specification and is the highest-priority structural improvement on the current prototype.

Keyed motor and module connectors throughout. One respondent in the Element J survey asked whether assembly was free-range or guided, and Run 3 of the Element I assembly test recorded a VelociMotor orientation reversal that required self-correction. Fully keyed connectors—physically asymmetric interfaces that can only mate in the correct orientation—would eliminate this error mode entirely and strengthen R1 compliance for younger assemblers in the 6–12 target age group, consistent with expert Jasmine Shields's recommendation that hardware interfaces should enforce correct assembly rather than relying on instruction compliance.

Surface finish improvement on 3D-printed parts. Two Element J respondents raised concerns about the visible layer lines and perceived fragility of 3D-printed components in a consumer product context. Applying a light sanding pass followed by a matte spray coat to exterior-facing chassis surfaces would significantly improve perceived quality without altering functional geometry. This is a low-cost, reversible improvement that directly addresses the aesthetic feedback received during the showcase.

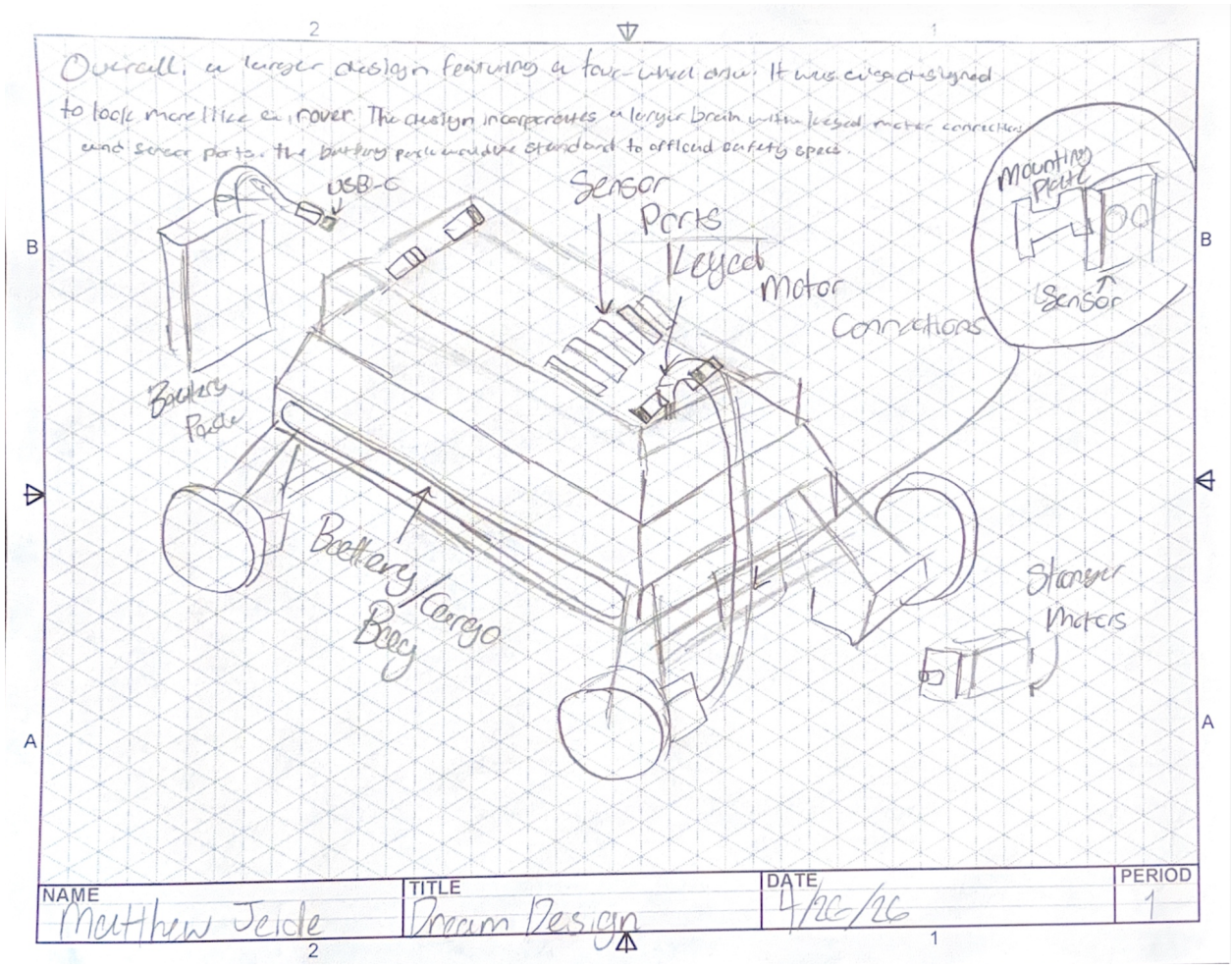
Line-following brain module implementation. The AGV firmware state and color sensor integration were architected in R3 but not physically realized due to time constraints. Completing the sensor wiring, calibrating the TCS34725 color sensor for black-on-white track detection, and validating the existing AGV firmware logic against a physical track would fulfill the 85% track completion design requirement (R5) and represent the single most significant functional expansion available within the current chassis architecture.

Packaging and product presentation design. An Element J respondent noted that packaging is "a very important part of selling the product." The M.O.J.O. team has developed brand identity assets—logo, typography in Bebas Neue and Inter, orange/black/white palette—but has not produced a physical packaging prototype. Designing and fabricating a kit box consistent with the established brand identity would complete the consumer-facing product presentation and address a gap identified directly by showcase attendees.

4 THE DREAM DESIGN

The M.O.J.O. dream design, illustrated in Figure L.1, represents a generational redesign of the platform rather than an incremental revision of R3. The core concept is a larger, four-wheel-drive rover with a chassis geometry and aesthetic inspired by planetary surface exploration vehicles—a direction that reinforces the STEM theme of the product while differentiating it visually from existing consumer toy competitors.

4.1 DREAM DESIGN ANNOTATED SKETCH



The dream design incorporates the following features and rationale:

Four-wheel drive drivetrain. R3 uses a front-wheel-drive differential configuration with a passive rear caster. The dream design replaces this with four independently driven wheels, each powered by a dedicated gearmotor. Four-wheel drive dramatically improves traction on varied surfaces—carpet, outdoor terrain, uneven classroom floors—and enables more complex behavioral programming, including skid-steer turning radius adjustment and individual wheel speed logging for odometry. This directly expands the engineering concepts accessible to the user, addressing the skill development dimension highlighted by Element J respondents who rated engineering concepts and gears/pulleys as the platform's strongest educational attributes.

Larger, rover-aesthetic chassis. The current R3 chassis is compact and functionally optimized but visually resembles a generic FDM prototype more than a finished product. The dream design features a wider, lower-profile chassis with overhanging wheel arches, a raised sensor deck, and a central cargo bay—proportions that evoke the Mars Science Laboratory and Perseverance rover silhouettes familiar to the target age group. One showcase respondent noted that "different options for the physical designs" would be desirable; a rover-aesthetic chassis provides a strong default visual identity while the modular brain system allows functional customization.

Stronger gearmotors. The N20 gearmotors in R3 are appropriate for the current chassis weight and surface conditions but have limited torque headroom for a heavier platform or varied terrain. The dream design would use higher-torque N20 or N30 variants—or purpose-selected DC gearmotors with metal gearboxes—selected based on the loaded chassis weight and target terrain specifications. This improvement directly supports the four-wheel-drive architecture and the more demanding operating environments a larger platform would encounter.

USB-C standard battery pack interface. R3 uses a custom-assembled 18650 cell with a TP4056 charging module, requiring the user to remove and externally charge the cell. The dream design replaces this with a standard consumer USB-C battery pack—housed in a dedicated cargo bay on the chassis underside—connected to the rover's power rail via a USB-C power delivery interface. This eliminates the custom charging circuit, leverages safety-certified off-the-shelf battery management, and allows the battery to be swapped or upgraded by the user without any electrical expertise. The USB-C interface also aligns with the R2 specification's requirement for standard 5V USB charging compatibility and directly addresses the teardown risk associated with the current cell installation process.

Keyed sensor ports with standardized mounting plates. The dream design features a row of keyed sensor slots along the top deck of the chassis, each accepting a standardized sensor mounting plate as illustrated in the detail bubble of Figure L.1. Each mounting plate carries a single sensor module—color, distance, IMU, or future sensor types—and connects to the brain via a keyed multi-pin interface that enforces correct orientation and prevents incorrect installation. This architecture makes sensor expansion genuinely modular and accessible to the target age group without requiring wiring knowledge, directly fulfilling the original co-play and inquiry-scaffolding design intent of the M.O.J.O. platform.

Custom-designed PCB brain. The most technically ambitious element of the dream design is replacing the current ESP32 development board and breadboard wiring with a purpose-designed printed circuit board integrating the microcontroller, motor driver, power regulation, and sensor headers into a single compact unit. A custom PCB eliminates the wiring fragility that contributes to assembly errors in R3, reduces the brain module's physical footprint, enables a cleaner enclosure design, and is a prerequisite for any future manufacturing scale-up. The PCB would be designed in KiCad, fabricated through a low-cost prototype PCB service, and documented with full schematic and layout files released under the project's open-source license—consistent with the open-source accessibility principle that 58.6% of Element J respondents identified as one of the platform's most appealing attributes.

5 CONCLUSION

The improvements identified for the R3 prototype are targeted, justified by empirical test data, and achievable within the existing chassis architecture. Battery module retention reinforcement and keyed connector implementation address the only formal test failure and the most common assembly error respectively, directly improving R3 and R5 specification compliance. Surface finish improvements and packaging design respond to aesthetic and consumer perception feedback gathered from 111 external evaluators during the Element J showcase. Line-following module implementation would complete the one unfinished functional requirement of the platform and represent the most educationally significant near-term expansion available without a chassis redesign.

The dream design represents a more ambitious vision: a larger, four-wheel-drive platform with a space-rover aesthetic, stronger motors, a standard USB-C battery interface, modular keyed sensor ports, and a custom PCB brain. Each of these features is grounded in a specific design rationale—stakeholder feedback, test results, or documented limitations of the current prototype—rather than arbitrary ambition. Together, they describe a platform that would more fully realize the M.O.J.O. project's original goals: a durable, accessible, educationally rich, and visually engaging rover kit that can grow with its users from basic assembly through advanced sensor programming. Both the near-term improvements and the dream design converge on the same underlying principle that has guided the M.O.J.O. project from Element A forward—that authentic STEM engagement requires a product that is honest about its engineering, accessible in its assembly, and ambitious in its educational potential.